# Emotion pipeline

GROUP 4

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- Business
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# Business

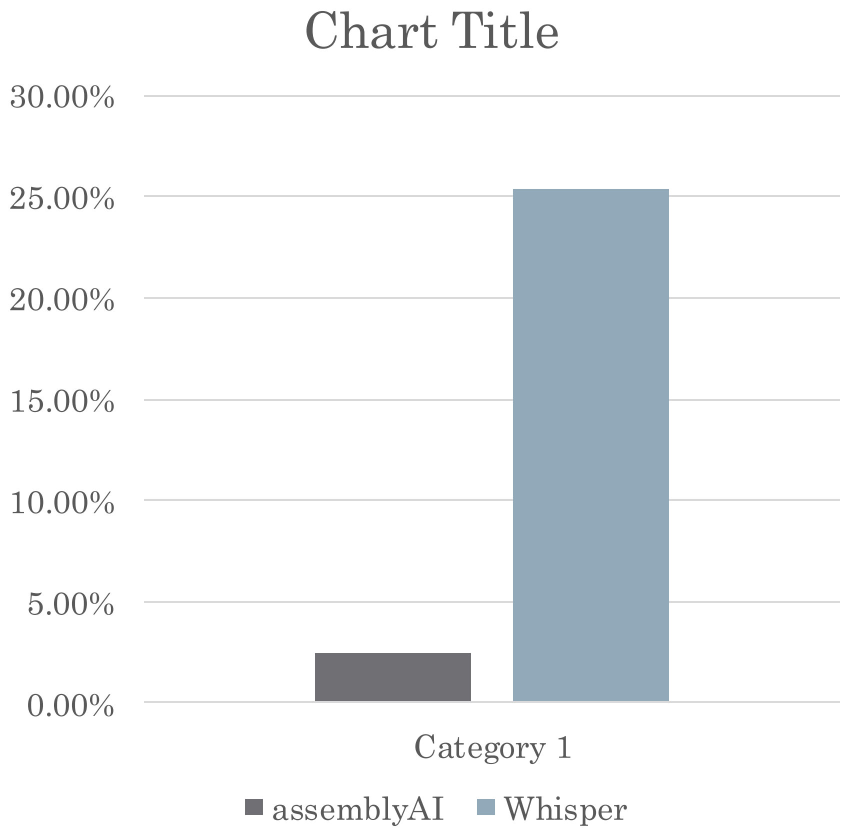
- Our understanding of the problem – LLM's
- The goal – cheaper and revolutionary
- Our project – New pipeline integrations

# The data

- Emotion analysis
- Go emotions (58k samples)
- Additional data added
- Machine translation
- Open subtitles NL - EN
- 200k sentences
- Problems
- Test data

# Speech to text

- Assembly AI
- Whisper
- Problems
- Tradeoff
- Results



# Features

TF-IDF

POS-Tagging

# Features

- Embeddings
- Speaker Tagging

# Models

| Name | Evaluation | Pro's | Con's |
|---|---|---|---|
| Logistic Regression | Weighted F1 0.46<br>Minority classes rarely classified | Fast & Easy | Doesn’t perform well on non-linear data |
| Naïve Bayes | Weighted F1 0.49<br>Completely failed to classify minority classes | Works well on small datasets | Doesn't capture relationship between words |
| LSTM | Weighted F1 0.55<br>Completely failed to classify minority classes | Word order<br>Long term dependencies | Prone to overfitting<br>Computationally heavy to train |

# Transformer

Pros:

- Contextual understanding

Cons:

- Computationally heavy to train
- Low explainability

• Roberta-base-go-emotions

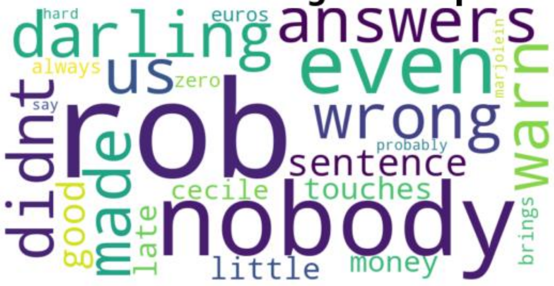
• Test Weighted F1 Score: 0.77

• Not able to effectively capture minority classes

# Transformer - Results

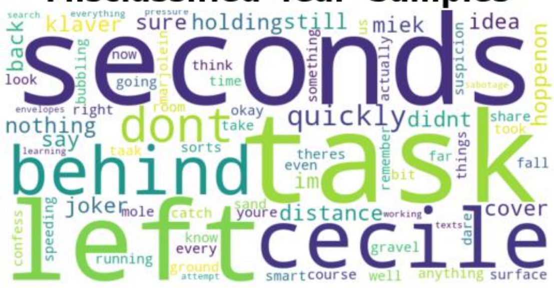
- Translation errors
- Short sentences contribute a lot of noise to the neutral
- Most confused classes Happiness and neutral

Misclassified 'anger' Samples




Misclassified 'disgust' Samples


Misclassified 'fear' Samples

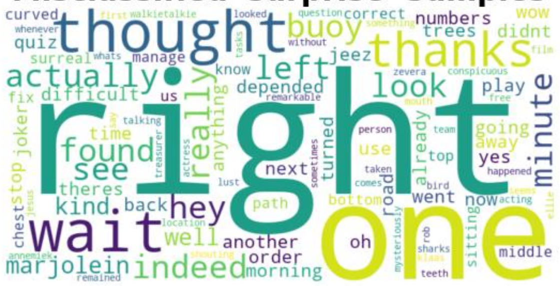



Misclassified 'happiness' Samples


Misclassified 'neutral' Samples


Misclassified 'sadness' Samples


Misclassified 'surprise' Samples



# Machine Translation

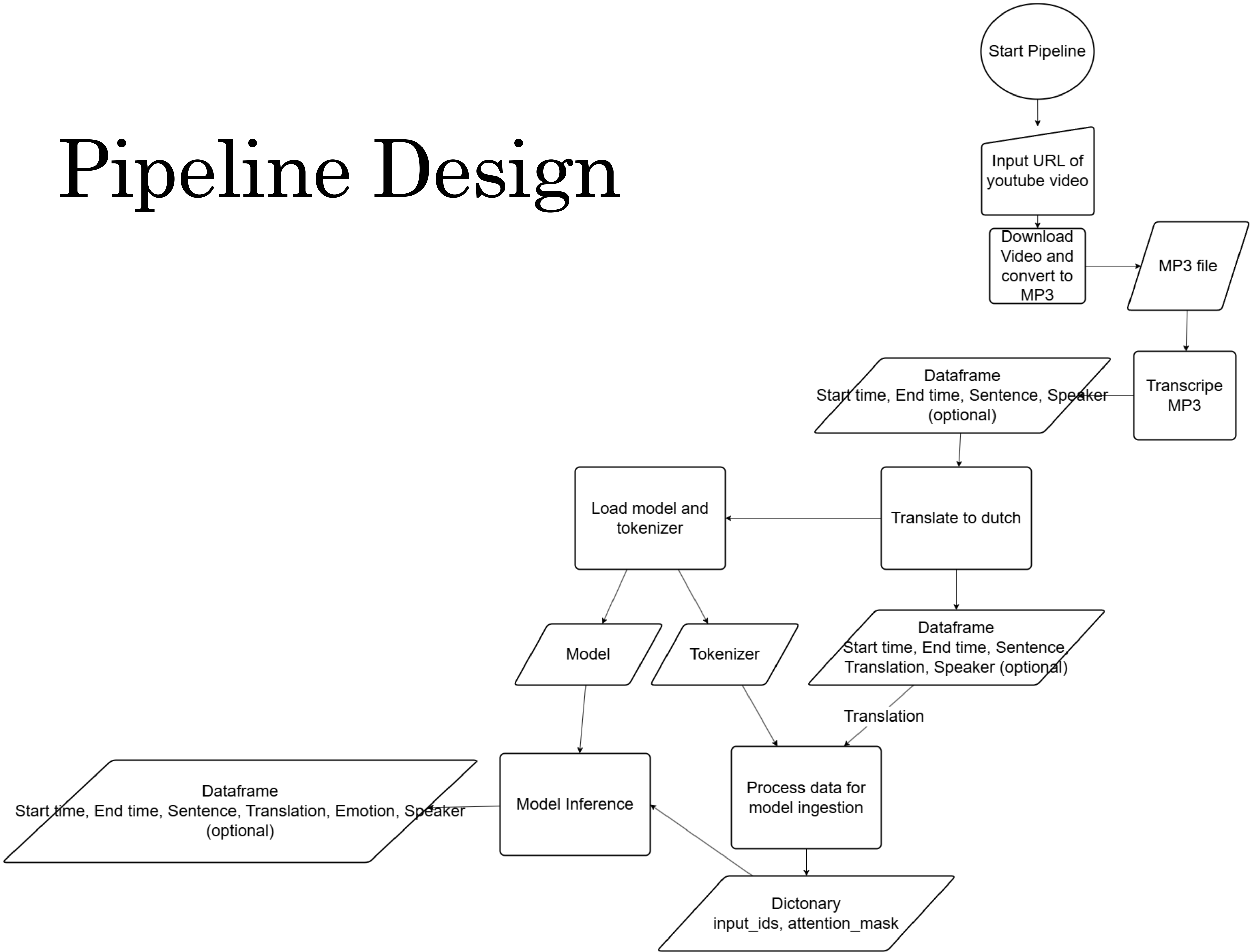
Data preprocessing and architecture (enc – dec)

Training

Results with checking and BLEU

Future direction in the pipeline

# Pipeline Design



# Pipeline Limitations

- Transcription accuracy VS Computional time
- Low recall and accuracy for minority classes
- Video download with only works on youtube
- Feature extraction not integrated

# Ethical considerations

Data privacy

Emotional profiling

Representation

# Future directions

- Data limitations
- Full integration
- Model optimization